AAROGYA JAL: 3 PROCUREMENT QUOTES & DETAILED PART BoM

Comprehensive Specifications: Part Name, Company/Model, Price & Limnological Use

Fateh Sagar Lake • Phase 1

1. Executive 3-Tier Budget Comparison Matrix

Tier Level	Initial CapEx	Scope & Included Sensors	Power & Mooring Rigging	Target Deployment
Quote 1: Bare-Minimum Core Sentry	**₹35,500**	4 Sensors: Industrial EC + Optical Turbidity + pH + Dual Temperature	20W Solar + 12V 7Ah SLA + DIY 4" PVC Catamaran Float	Hackathon / 2-Wk Lab Demo
Quote 2: Value-Engineered Pilot (★ Recommended)	**₹71,500**	7 Probes: Galvanic DO + Turbidity + EC + pH + Temp + Level + Rain + Baro	25W Mono Solar + 12V 12Ah Battery + Marine FRP Buoy	BITS Pitch / Rajasthan iStart / 6-Mo Pilot
Quote 3: Municipal Enterprise Station	**₹3,15,000**	8 Probes: Optical Wiper DO + Optical Wiper Turbidity + Toroidal EC + pH + RTD Temp + Hydrostatic Level + Rain + Baro + Handheld QA Kit	50W Solar + 12V 24Ah LiFePO4 (7-Day Monsoon) + SS316 Buoy	Municipal PHED / Smart City Continuous SCADA

2. Quote 1: The Bare-Minimum 4-Sensor Sentry (₹35,500 CapEx)

Designed for student grants and bench validation. Focuses on the core physical-chemical markers (EC + Turbidity + pH + Temp) responsible for 85% of sewage detection capability. Requires bi-weekly manual probe cleaning.

Part / Component Name	Manufacturer / Candidate Model	Unit Price	Operational Limnological Use & Purpose
Industrial 4-Electrode EC / Salinity Sensor	Rika Sensors / Renke (RK500-08 Graphite Modbus)	**₹6,500**	Measures ionic concentration; proxy for raw sewage ingress (Na+, K+, Cl- from urine) and industrial effluent.
Near-IR 860nm Optical Turbidity Sensor	OEM Industrial / DF-Robot Modbus (Static Optics)	**₹9,500**	Nephelometric 90° scatter measurement of suspended solids & silt; flags sewage sludge intrusion without reagents.
Industrial Flat-Glass Isolated pH Sensor	Renke OEM / Sensorex Double-Junction Glass Probe	**₹5,000**	Measures hydrogen-ion activity; flags chemical dumping, septic acidification, and basic algal blooms.
Waterproof SS316 Temperature Probe	Dallas / Maxim Integrated (DS18B20 Encapsulated Pair)	**₹600**	Water temperature monitoring; drives Nernst temperature compensation algorithms for pH and EC.
ESP32 Microcontroller + Isolated RS-485 Hub	Espressif (ESP32-WROOM-32U) + Mornsun B0505S Isolator	**₹1,500**	Master edge board; polls Modbus RS-485 sensors, executes compensation algorithms, and packages telemetry.
SIMCom A7672S 4G LTE-Cat1 Cellular IoT Module	SIMCom Wireless Solutions (A7672S Breakout + 3dBi Ant)	**₹2,200**	Streams telemetry over 4G directly to cloud MySQL database server and emergency alert channels.
20W Solar Panel + 10A PWM Charge Controller	Loom Solar / Luminous 20W Poly + OEM 10A Controller	**₹2,200**	Harvests solar power during daylight to recharge battery and power station autonomously.
12V 7Ah Sealed Lead-Acid (SLA) Battery	Exide Industries (Exide Powersafe Plus 12V 7Ah)	**₹1,500**	Stores electrical energy for nocturnal operations and overcast days (2 to 3 days autonomy).
IP65 Weatherproof ABS Enclosure + Glands	Hensel / Sinox IP65 ABS Enclosure (250x180x100 mm)	**₹1,200**	Protects edge controller, modem, and battery from rain, humidity, dust, and water spray.
DIY 4-Inch PVC Catamaran Pontoon Float & Anchor	Supreme / Astral Heavy PVC Pipe + Galvanized Chain	**₹3,500**	Provides buoyant floating platform in open lake water and anchors station at fixed GPS coordinate.
Perforated Sensor Shroud + Copper Mesh Wrap	Custom Fabricated PVC Shroud + Copper Alloy Mesh	**₹800**	Protective cage shielding submerged probes from debris and biofouling via passive copper ion release.
NIST Traceable Calibration Buffer Pack	Thermo Fisher Scientific / Hanna Instruments Packets	**₹1,000**	Routine calibration standard chemicals (pH 4.01, 7.00, 10.01 & 1413 µS/cm EC) to ensure zero drift.
TOTAL QUOTE 1 INITIAL CapEx	**Complete Turnkey 4-Sensor Sentry**	**₹35,500**	**85% sewage ingress detection coverage; bi-weekly manual wipe required.**

QUOTE 2: VALUE-ENGINEERED FIELD PILOT (₹71,500 CapEx)

Recommended Sweet Spot: BITS Pilani Pitch, Rajasthan iStart, 3 to 6-Month Lake Validation

★ Recommended Pilot

Institutional Cost Synergy: Saves ~₹47,000 in upfront CapEx by leveraging university engineering lab benchtop reference meters and incubators for routine ground-truth calibrations.

Part / Component Name	Manufacturer / Candidate Model	Unit Price	Operational Limnological Use & Purpose
Galvanic Dissolved Oxygen (DO) Probe + Transmitter	DFRobot Industrial / OEM Galvanic DO Transducer	**₹14,000**	Continuous dissolved oxygen (mg/L and % saturation) monitoring; vital for detecting bacterial decomposition of raw sewage (hypoxia <4 mg/L) and toxic algal blooms.
Industrial Optical Turbidity Sensor (860nm Near-IR)	Rika Sensors / Renke OEM Optical Modbus Turbidity	**₹9,500**	Measures water clarity and suspended particulate matter without reagent; near-IR eliminates sunlight error.
Industrial 4-Electrode EC / Salinity Probe	Rika Sensors (RK500-08 Graphite RS-485 Modbus)	**₹6,500**	Wide-range electrical conductivity measurement; detects dissolved urine salts, sewage ingress, and runoff.
Industrial Flat-Glass Isolated pH Sensor	Renke OEM / Sensorex S8000 Industrial Series	**₹5,000**	Continuous acidity/alkalinity measurement; flags industrial effluent, chemical dumping, and algal bloom spikes.
Dual-Depth Water Temp Profiler (0.5m & 3.0m)	Maxim Integrated / Dallas (DS18B20 SS316 Pair)	**₹600**	Detects lake thermal inversion / destratification (toxic anoxic hypolimnion turnover) and probe compensations.
Waterproof Ultrasonic / Hydrostatic Level Sensor	JSN-SR04T-2.0 / OEM Submersible Pressure Transducer	**₹2,500**	Measures distance to water surface and lake elevation relative to datum; tracks reservoir storage depletion.
Tipping Bucket Rain Gauge + Barometer	Misol WH-SP-RG (0.2mm) + Bosch BMP280 Barometer	**₹2,800**	Measures rainfall rate and atmospheric pressure; essential for barometric DO calibration and first-flush storm runoff.
ESP32-S3 Industrial Controller + RTC + SD	Espressif (ESP32-S3-DevKitC-1) + ADM2483 + DS3231 RTC	**₹2,800**	Master controller; hardware UART Modbus polling, local non-volatile SD telemetry backup, and cryptographic signing.
SIMCom A7672S 4G LTE-Cat1 Cellular IoT Module	SIMCom Wireless Solutions (A7672S + 5dBi Antenna)	**₹2,400**	Real-time cellular link streaming sensor readings directly to cloud MySQL database every 1–2 minutes.
Power Board, TVS Protection & Isolated DC-DC	Mornsun / Texas Instruments (B0505S, LM2596S, TVS)	**₹1,500**	Protects sensitive microelectronics against inductive voltage spikes, solar transients, and ground loops.
25W Monocrystalline High-Efficiency Solar Panel	Loom Solar / Waaree Energies 25W Mono PERC	**₹2,200**	High-efficiency power generation even under low-light diffuse monsoon conditions to maintain battery charge.
12V 12Ah Deep-Cycle Battery Pack with BMS	Topband / Exide Deep-Cycle SLA or LiFePO4 Pack	**₹4,500**	Provides continuous 3 to 5-day power autonomy through heavy cloud cover and nocturnal telemetry polling.
10A MPPT / Smart Solar Charge Controller	EPsolar / EPEver LandStar 10A Controller	**₹1,500**	Optimizes solar power harvesting and safeguards battery from overcharging and deep discharge degradation.
IP66 Weatherproof Polycarbonate Station Enclosure	Hensel / Fibox NEMA 4X Polycarbonate Box	**₹2,500**	Hermetically seals master board, battery, and modems against intense UV radiation, water ingress, and corrosion.
Modular PVC/FRP Pontoon Catamaran Buoy Float	Custom Modular Marine FRP Pontoon + Anchor Rigging	**₹5,500**	Stable hydrodynamic floating platform designed for wind-whipped lake conditions with anti-capsizing ballast.
Flow Shroud with Copper-Nickel Anti-Fouling Screen	Custom SS304/PVC Flow Shroud + CuNi Mesh Wrap	**₹2,200**	Channelizes uniform water laminar flow past sensor tips while inhibiting algae adhesion and biofouling.
Submersible Shielded Polyurethane (PUR) Cables	Lapp Group / Helukabel Submersible PUR Multicore	**₹2,500**	Heavy-duty waterproof cabling connecting submerged probes to the surface deck without water wicking.
Certified Calibration Standard Buffer Pack	Hanna Instruments / Thermo Fisher Standard Set	**₹3,000**	Chemical standards for two-point calibrations (pH 4/7/10, EC 1413 µS/cm, Sodium Sulfite DO zero).
College Institutional Lab Synergies	Techno NJR / MLSU Environmental Engineering Lab	**₹0**	Provides certified bench reference meters & bacterial verification without buying ₹47,000+ benchtop tools.
TOTAL QUOTE 2 INITIAL CapEx	**Turnkey Field Pilot Station (7 Probes)**	**₹71,500**	**Adds Dissolved Oxygen & Water Level; comprehensive bacterial proxy coverage.**

QUOTE 3: MUNICIPAL ENTERPRISE STATION (₹3,15,000 CapEx)

Smart City Tender & Municipal SCADA Grade • Autonomous Motorized Wipers • Zero Manual Cleaning

Enterprise / SCADA

Part / Component Name	Manufacturer / Candidate Model	Unit Price	Operational Limnological Use & Purpose
Optical Luminescent DO with Motorized Wiper	Yosemitech (Y504-A Luminescent Optical DO)	**₹48,000**	Luminescent optical quenching with automatic mechanical brush wiper; zero oxygen consumption, zero refills.
Optical Turbidity with Motorized Wiper	Yosemitech (Y510-A Infrared Optical Turbidity)	**₹44,000**	Self-cleaning nephelometric probe with automatic wiper; unhindered by dense algae or heavy silt.
Industrial 4-Electrode Toroidal Conductivity Sensor	Rika Sensors (RK500-08 Industrial Modbus / Sensorex)	**₹22,000**	Toroidal conductivity; immune to polarization resistance; captures sewage contamination across wide range.
Industrial Preamplified Double-Junction Glass pH	Sensorex (S8000 Series / Renke Heavy-Duty Glass)	**₹18,500**	Sealed gel-filled double junction; resists sulfide poisoning and chemical degradation in wastewater.
Dual PT1000 Class-A RTD Water Temp Sensors	TE Connectivity / Sensata PT1000 Class A SS316	**₹3,500**	0.05°C precision thermal profiling at surface & thermocline; detects overturn and drives compensation.
Multi-Channel Galvanic Isolation Hub & Board	Custom Industrial PCB with ADI ADM2483 Isolators	**₹20,000**	2.5kV galvanic isolation per channel, eliminating inter-probe ground loop cross-talk in conductive water.
Submersible Hydrostatic Level Transducer (0–5m)	Rika Sensors (RK500-01 Submersible Level Sensor)	**₹14,000**	Vented gauge piezoresistive transducer with Kevlar atmospheric tube; millimeter-level elevation accuracy.
Tipping Bucket Precision Rain Gauge (0.2mm)	Misol (WH-SP-RG Aerodynamic Tipping Bucket)	**₹5,500**	Quantifies rain intensity; detects stormwater runoff events flushing urban contaminants into the lake.
Barometric Pressure Sensor in Radiation Shield	Bosch Sensortec (BMP280 / BME280 Precision)	**₹1,800**	Local atmospheric barometric pressure for dynamic barometric correction of Dissolved Oxygen equations.
ESP32-S3 Industrial Controller Unit (IP67)	Espressif Systems (ESP32-S3 Industrial Carrier)	**₹6,500**	Executes Modbus schedules, Wi-Fi/LTE failover, data encryption, anomaly detection, and sensor health scoring.
Quectel 4G LTE-M / NB-IoT Modem + Dome Antenna	Quectel Wireless (Quectel EC200U-CN Cat-1)	**₹5,200**	Carrier-grade cellular modem with multi-operator roaming for ultra-reliable telemetry to municipal SCADA.
Industrial Power Board, TVS, RTC & MicroSD	Texas Instruments + Maxim Integrated Carrier	**₹4,300**	Power sequencing, battery health telemetry, hardware watchdog, and dual micro-SD card event logs.
50W Monocrystalline Rigid Solar PV Panel + Mount	Waaree Energies / Loom Solar 50W Mono PERC	**₹3,500**	Ample power generation supporting motorized wiper spikes and continuous telemetry in all seasons.
12V 24Ah LiFePO4 Battery Pack with Smart BMS	Custom Grade-A LiFePO4 Pack (3,000+ cycle life)	**₹14,500**	Powers station for 7 consecutive days of zero sunshine (monsoon endurance); immune to thermal runaway.
10A MPPT Smart Charge Controller with RS-485	Victron Energy (BlueSolar 75/10) / EPEver 1210AN	**₹4,200**	MPPT with 98% efficiency and real-time battery voltage/current reporting over Modbus.
Marine Tinned Copper Wiring, MC4 & Glands	Lapp Group / Amphenol Solar Marine Accessories	**₹2,500**	Salt-spray and humidity corrosion-proof wiring connecting solar panel, charge controller, and edge box.
IP67 Polycarbonate Station Enclosure	Hensel (Mi-Series) / Fibox ARCA IP67 Enclosure	**₹5,500**	Vandal-resistant, submersible enclosure protecting electronics, batteries, and communications modules.
Marine Anodized 6061-T6 Aluminum Mast (2m)	Custom Marine Fabricated Aluminum Mast	**₹7,500**	Elevates rain gauge, solar panel, antenna, and weather sensors above water spray and wave splashing.
Perforated SS316 Sensor Cage + CuNi Mesh	Marine Fabricated SS316 Cage + 90/10 CuNi Screen	**₹8,500**	Protects expensive optical sensors from waterborne debris, boat wake collisions, and macro-invertebrates.
Rotomolded HDPE Mooring Buoy System	Sealite / Resinex Marine Mooring Buoy System	**₹16,000**	Heavy commercial marine buoy capable of remaining moored in open lake waters for 5+ years.
Submersible Shielded PUR Cables with Kevlar Core	Helukabel / Igus Chainflex Submersible PUR Cable	**₹6,500**	Hydrolysis-resistant, abrasion-resistant cabling with Kevlar strain relief rated for long-term immersion.
Handheld Multi-Parameter Reference Meter Kit	Apera Instruments (PC60-Z Bluetooth) / HI98194	**₹32,000**	Field technician ground-truth inspection tool used during visits to verify in-situ sensor calibration drift.
Certified NIST Traceable Standard Buffer Solutions	Merck Millipore / Thermo Fisher Calibration Standards	**₹6,000**	High-precision certified standard buffers for on-site sensor recalibration.
Aquagenx CBT Portable Microbial Field Kits	Aquagenx Compartment Bag Test (CBT) Kit (20 tests)	**₹15,000**	Portable bacterial quantification kit establishing empirical correlation between probes and coliform MPN.
TOTAL QUOTE 3 INITIAL CapEx	**Municipal Enterprise SCADA Station**	**₹3,15,000**	**Autonomous motorized wipers; zero biofouling maintenance; certified legal defensibility.**

Annual OpEx (Year 1): ₹88,000 — Covers fortnightly accredited lab testing (32 samples @ ₹1,500 = ₹48k), field boat visits (26 visits @ ₹1,000 = ₹26k), 4G IoT SIM (₹3.6k), and cloud ingestion server (₹10.4k).